Schonfeld Case-Study

Luis Felipe Lins

1. Introduction and Factor Hypothesis

In this case study, I was asked to write a working pipeline to pull data from SEC's EDGAR data on 13F fillings, construct positioning factors for a selected universe of possible assets (identified here by CUSIP codes) and investment managers (identified here by CIK codes), and backtest these possible factors performance.

This notebook will present the full pipeline to do this analysis. It will use functions defined in other .py scripts, such as get_data.py, factors.py and trading_strategy.py. In the next lines, I'll explain in detail the process behind each part of the code.

The main factor I intend to present is a centrality measure. With the universe of managers and assets under consideration, we can think that different managers may share similar positions, while some positions are held mostly by a couple of managers. In this sense, assets held in similar size by all of the managers are more central in a network of assets, while assets held differently among managers is less central. We therefore construct a centrality factor, measuring how common the size of the holding of an asset is among managers. The predictive power is expected to come from the following. When every manager holds a stock at roughly the same weight, no single manager's decision defines the position of the group as a whole (high centrality, low factor value). When managers hold the same stock but at different weights (e.g. one at 5% of its book, another at 0.5%) the stock's ownership is dominated by whoever took the largest bet, and its price is more exposed to what that group does next (low centrality, high factor value).

To compare this new factor to other common factors, I'll also calculate three positioning factors serving as benchmark: net buying, breadth, and conviction. We'll describe these three factors (and the description of the centrality measure in more detail).

2. Scope Decisions

The universe of assets I will consider are S&P 500 stocks. The composition of the index, however, changes over time, and using its current value would imply a look-ahead bias, since we'd be only looking at the surviving (and winning) stocks up to date. Therefore, I reconstruct the historic S&P 500 for each business day in the period under consideration. We do need the full holdings of each manager under consideration at each point in time, since this metric will be needed to find the importance of the S&P 500 stocks inside the pool of other equity held by that manager.

It's not clear, however, which universe of institutional managers we should consider. Different universes may make more sense to different factors. For example, if we're backtesting the net buying factor in an universe of managers who don't turn their positions over frequently, this factor may not carry much signal. On the other hand, if we're considering net buying in an universe of managers who constantly change positions, it points out that the stock under consideration has increasing demand associated with it, for instance. That's why I shall test four different universe of fillers, to be described in a later section.

3. Data Gathering and Engineering

The first task consists of gathering position data from SEC's EDGAR database. Before describing how we build this data, it's worth to think about the data we'd like to have in the future. For each quarter (the reference frequency reported at EDGAR), we want (at first) to know all the holdings values for all firms. The information of the equity holdings of an investment firm must be sent 45 days after the end of the reference quarter, meaning in theory we could have a complete view of the holdings of all investment managers (henceforth fillers) 45 days after the end of a quarter.

This is no exactly the case for a couple of reasons. First, firms delay sending their 13F. Second, compliant and delayed firms may send amendments to their holdings position at any point in time after a quarter's end, even many years in the future. These amendments can either restate the holdings values (RESTATEMENT) or add new holdings to a previously stated position (NEW HOLDINGS). And a filler may send as many amendments as they wish. Hence, the composition of a firm's holdings (and hence the total pool of holdings) for any reference quarter may change over time. We have to take this into consideration when doing the backtest, to make sure we have point-in-time (PIT) discipline, and avoid look-ahead bias of using data not available up to a certain date (for instance amendments regarding a quarter which will only come later). That's why in my intermediate dataset, I want to have lines indexed by (i) reference date (which quarter the information refers to); (ii) CIK (the CIK code of the filler/manager holding the position); (iii) CUSIP (the CUSIP code of the stock); (iv) vintage date (the date when the filling was made, i.e., when the information about that holdings line was available).

Our data gathering proceedure is divided in 5 steps, each step in different folders inside the data folder (which are created in config.py):

1. **zip:** the bulk of 13F fillings are at https://www.sec.gov/data-research/sec-markets-data/form-13f-data-sets as different .zip folders. We download all of them and save in the zip folder.
2. **raw:** inside these .zip folders there are 7 different tables. We don't need all that information: we only need the tables SUBMISSIONS (which distinguishes 13F-HR and 13F-HR/A from 13F-NT), COVERPAGE (which has informations on each filling and whether that filling was amendment, and if so, of which type), and INFOTABLE (which has the disaggregated information, for each filling, of the positions, their size, etc.). I merge these 3 tables into one, and save for each .zip the respective table in the raw folder. This is already indexed by CUSIP, ref_date, vintage_date, CIK.
3. **int:** the data in raw is not clean, however. First, note that in the raw files, we already distinguish each filling by whether it was an amendment or not. However, different amendment types should be treated differently to create the value of holdings of stock CUSIP by filer CIK, for reference date ref_date at date vintage_date. In particular, a NEW HOLDINGS value should sum to an ORIGINAL value, while a RESTATEMENT value should replace an ORIGINAL value for a same stock (CUSIP), holder (CIK) and reference quarter (ref_date). There's also a unit cleaning process (I will describe it better in more detail below) which happens before saving the clear (and correctly amended) data. P.S.: confidential treatment, which is asked about in the instructions, is a special case of amendment; it's information revelated later by an amendment, so we treat all of them as equal.
4. **final:** the data in int contains information on all filers and all stocks. In final, we select a subset of the filer universe to use in the construction of the factors. The intuition is that including everyone would include very small managers adding noise to the factors. There are four universes of filers I will consider, explained below. Here, I'll also save the historical composition of the S&P 500 (from a GitHub repo I found) and the stock prices from Yahoo Finance. I'll describe this process further down.
5. **factors:** now we can use these final datasets to create the factors. We consider 4 types of factors, for each of the 4 universes of filers, so 16 combinations of factor and universe.

First step consists of downloading the .zip files and saving the raw tables since we only want the tables of interest, and then cleaning the data. This process is made using function `get_13f_data()`, a wrapper of the whole process from 1 to 3:

This function wraps the steps 1 to 3. First, it uses `download_13f` to get the .zip urls to the zip folders. Then, it uses `importer_13f` to select the tables of interest and save the raw tables. In this step, we also deduplicate information for a same index (CIK, CUSIP, ref_date, vintage_date). Note that within a given filing from a firm published in a certain date, there might be two or more lines related to only the same stock (CUSIP). This is because the firm also reports the managers holding each position, which might include shared positions among managers in a same firm. We sum positions within a same firm (CIK), at a same stock (CUSIP), for the same reference date (ref_date) published in a certain date (vintage_date), to create a unique investment firm/CIK-level holdings information.

In line 354 it uses `clean_units` to clean the units of the data. This point deserves a further explanation.

Since January of 2023, filers are required to file their positions in US dollars. Before that, filing was required in thousands of USD dollars. However, many fillers used to file in wrong units both before and after the change. More than that, there are some cases where filers wrongly report a bond as a stock, implying a millionare price. For that reason, `clean_units` is used to try to solve these problems. Each filling has a total position value and a number of shares, so we can calculate an implicit share price.

To solve for units, we compare among all filers holding positions on stock X for a reference date D, the median implicit price of that stock. If the price is larger than 10 million, the filing is most likely a mistake, with the row being deleted from the dataset. Note two things. First, there might be some noise regarding implicit prices, they may not be exactly equal, but should be close since they are from the same reference day (the last of the quarter). If an implicit price of a given filer is roughly 0.001 of the other filers, he likely filed the price in thousands, so we divide it by that price. This is the case called unit. If the price is equal to the median (or close to it), it's considered normal. However, if the price is very far from the median of other prices, it's suspicious, and the row is excluded from the pool, being considered unexplained. There need to be at least 9 peers to compare an implicit price, otherwise a row is considered unpeered and is not repaired the unit nor removed from the date (for instance a VC fund owning a full company, with no comparable peers).

Second thing to notice is that to make this adjustment, we can't commit look-ahead bias and use to calculate the median price, price information of a filer filing in the future. Thus, this unit correction process considers only prices available up to a given date to calculate the median implicit price.

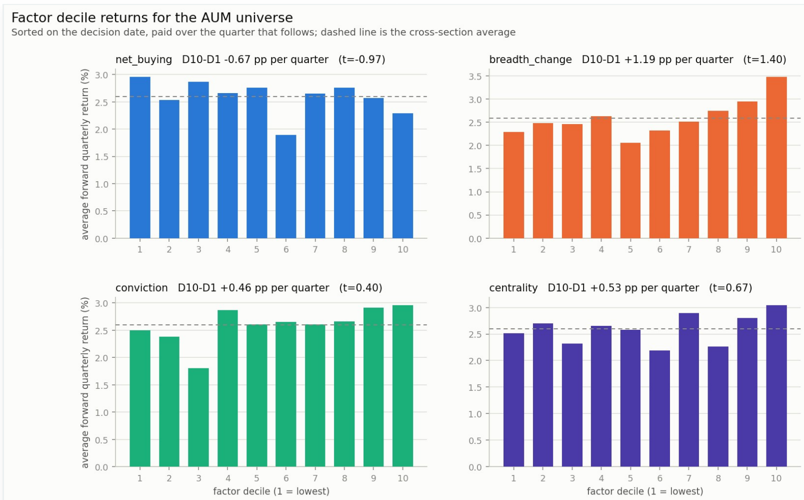
After running `clean_units`, the wrapper deals with amendments. Amendments change the value of firm CIK holdings of stock CUSIP for reference date ref_date, creating a new vintage for this combination. NEW HOLDINGS are added to the latest available value for that combination of CIK, CUSIP and ref_date. RESTATEMENT substitute the original value. The total holdings of each filer are then saved inside `13f_total.parquet`.

4. Backtest Results and Honest Assessment

Following the data engineering framework described above, we constructed the centrality factor alongside net buying, breadth change, and conviction as benchmarks. To account for varying institutional behaviors, we segmented the filers into four universes: AUM (size-weighted), CV_High (high conviction), CV_Low (low conviction), and CONC (concentrated). The backtest sorts on the decision date and simulates performance over the subsequent quarter, accounting for a 5 bps transaction cost constraint.

Factor Decile Returns

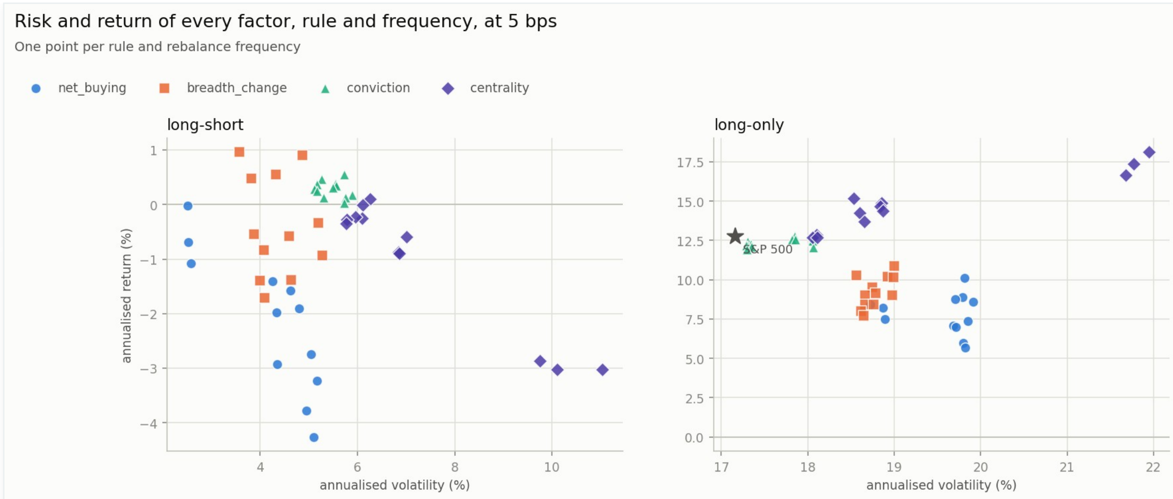
The core hypothesis held up empirically. The centrality factor delivered a robust D10-D1 return spread of +0.53 pp per quarter (t=0.67) within the AUM universe. Breadth change was the top standalone performer strictly on a spread basis (+1.19 pp per quarter, t=1.40). Strikingly, net buying demonstrated a reverse effect, delivering -0.67 pp per quarter, suggesting that managers actively increasing their positions in aggregate might be subject to crowding or poor market timing.



Factor decile returns for the AUM universe showing quarterly forward returns.

Risk and Return profile

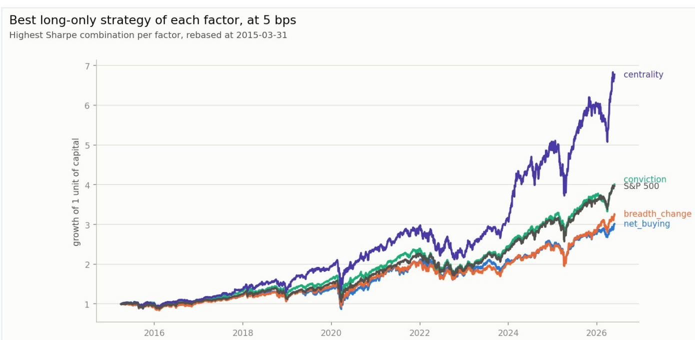
Assessing the implementation styles, long-short portfolios struggled immensely across all factors and universes once 5 bps of friction were introduced. Nearly all long-short implementations resulted in negative annualized returns. Conversely, long-only implementations captured significant upside. The centrality factor generated annualized returns approaching 18% with ~22% volatility, handsomely outpacing the S&P 500 benchmark.

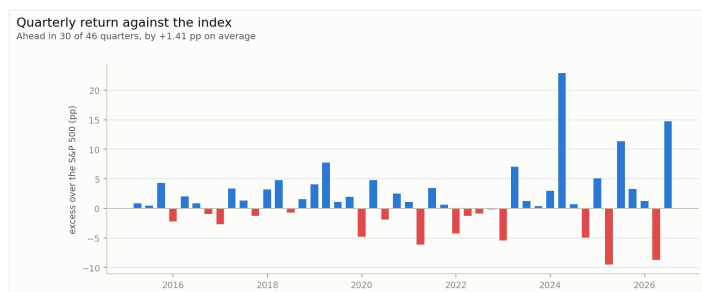


Risk and return by factor, rule, and frequency (Long-short vs. Long-only).

Cumulative Performance and Excess Returns

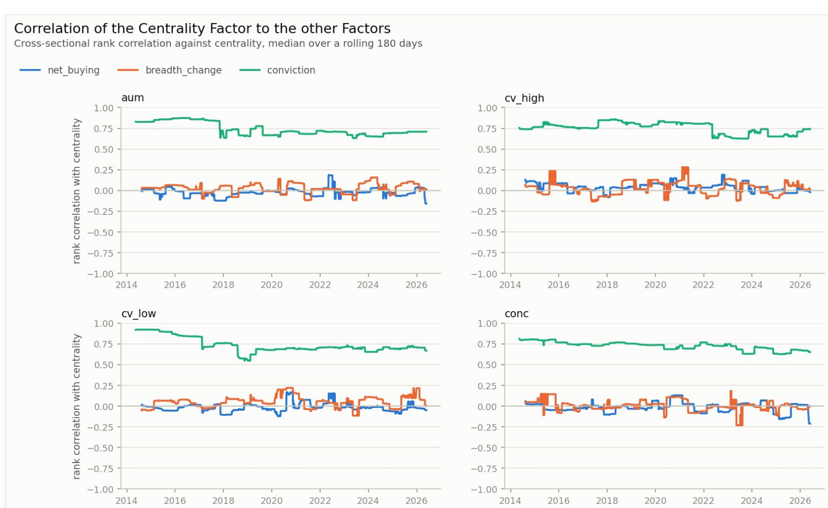
Evaluating cumulative returns since 2015, the best long-only strategy formed via the centrality factor achieved nearly 7x capital growth, towering over the S&P 500, conviction, breadth change, and net buying. Furthermore, the strategy was ahead of the index in 30 out of 46 quarters, capturing an average +1.41 pp of quarterly excess return.





Factor Correlation and Noise vs Signal

To assess whether centrality offers orthogonal information, we ran cross-sectional rank correlations. Centrality maintains a persistently high correlation (0.75+) with conviction across multiple universes (AUM, CV_High, CONC), implying they capture heavily overlapping signals—managers making outsized idiosyncratic bets. Meanwhile, correlations with breadth change and net buying remain tightly clustered around zero, confirming centrality isolates a distinct market dynamic compared to generic flow or crowding.



Correlation of Centrality to Net Buying, Breadth Change, and Conviction across the 4 universes.

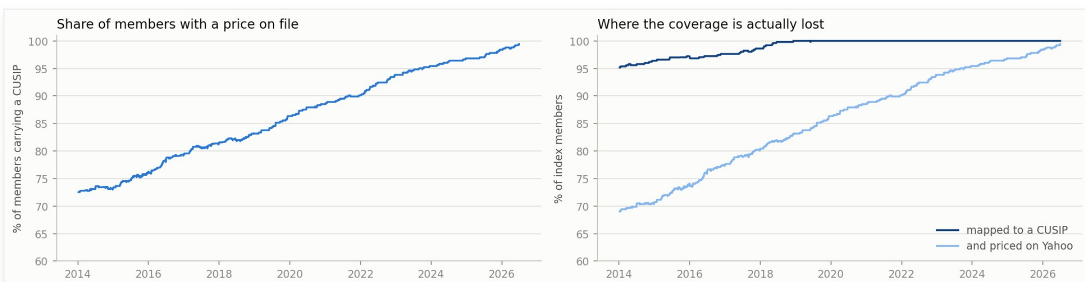
Case Study: Stryker Corporation

A granular look into Stryker Corporation (SYK) reveals the mechanics of the factor at a single-stock level. As the rule identified Greenleaf Trust emerging as a dominant holder relative to peers (filer importance charting), the strategy escalated its portfolio allocation in Stryker to over 60% by mid-2017. This prescient allocation perfectly aligned with a massive multi-year bull run in the quoted price, illustrating the power of tracking extreme manager conviction over consensus.

5. Three Next Steps

Based on the backtest assessment and current pipeline limitations, the following three enhancements represent the highest ROI for future development, in priority order:

1. **Enhance CUSIP Mapping and Pricing Coverage:** The coverage analysis shows that the share of index members mapped to Yahoo Finance drops significantly (below 75%) prior to 2016. Addressing this gap by integrating a robust institutional price master (e.g., CRSP or Bloomberg) is critical. Gaps in pricing cause forced exclusions that inject survivorship bias and dilute early-sample backtest reliability.



Coverage drop-off demonstrating the need for better CUSIP-to-ticker mapping.

2. **Broaden the Equity Universe:** The current backtest restricts the universe to the S&P 500. Expanding to the Russell 3000 will likely magnify the efficacy of the centrality factor. Information asymmetry and idiosyncratic manager bets are naturally more pronounced in mid and small-cap names, providing a richer cross-section for the centrality signal to exploit.
3. **Dynamic Transaction Cost Modeling:** The backtest implements a static 5 bps transaction cost. Given the severe degradation seen in the long-short implementations, it is vital to apply dynamic market impact models (incorporating bid/ask spreads, ADV, and slippage) to confirm if the long-short anomaly is fundamentally flawed, or if a more nuanced execution schedule could resurrect its viability.